Spatial Instruction Following in World–Action Models

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Abstract: When a robot succeeds at right of the bowl but fails at left of the bowl, does it misunderstand the instruction, or is one placement harder in that scene? We study Cosmos 3 Nano and DreamZero-s2, a custom action-guidance configuration, in a simulated cube-placement task. For each model, 108 trials cross two commands with two object layouts while keeping the model, robot, cameras and controller fixed. Mirroring the objects changes DreamZero-s2 completion from 13/54 to 50/54. Most original-layout failures occur at pickup or transport. Nano remains highly successful, but which side it reaches farther reverses. At its goal-dependent endpoints, every matched pair has the requested LEFT/RIGHT ordering; a pretermination check preserves ordering but reduces separation. Thus scene layout can change task performance even when a model responds to the directional words.

Keywords: World–action models, spatial instructions, robot manipulation

1 Introduction

A robot asked to place a cube beside a bowl must recognize left and right, grasp, transport and release. The instruction can redirect motion correctly while placement still fails. Different LEFT/RIGHT success rates therefore do not show whether language failed to influence the policy or execution failed to complete the task.

This distinction matters for world–action models. DreamZero jointly models future video and actions, while Cosmos 3 supports world modeling and policy generation [1, 2]. Spatial following must still be established through robot behavior; world-model benchmarks likewise assess functional utility alongside visual quality [3].

We mirror movable-object positions while fixing the policy and commands. LEFT and RIGHT trials begin from the same state in each layout. We separately measure the final cube–bowl relation, requested-side depth and task success. DreamZero-s2 completion changes substantially with layout; Nano placement changes near a success ceiling. Task success and directional response can therefore support different conclusions.

2 Experimental Setup

Task and models. We evaluate Cosmos 3 Nano and DreamZero-s2 on a Rubik's-cube placement task in the DROID/RoboLab simulator. DreamZero-s2 is the DreamZero-DROID checkpoint with custom action-guidance scale 2; the baseline scale 1 is not used. The command is Put the Rubik's cube to the [left/right] of the bowl. It stays fixed for at most 450 actions, with no intermediate hints.

Scene intervention and repeated trials. The reflected layout mirrors the centers of all movable objects across the robot's forward midline. The robot, cameras, fixed scene geometry and controller remain unchanged. LEFT continues to mean the robot's left in both layouts (Figure 1).



Figure 1: Change the layout, repeat the commands. Both LEFT and RIGHT instructions are tested in each arrangement. Movable objects are reflected, while the robot and cameras remain fixed. The diagram is schematic; it does not show measured coordinates or a recorded trajectory.

Each model has 27 matched blocks. Each block runs both commands in the original and reflected layouts, giving 108 trials per model. Repeats within a layout share one starting physical state: these are two layouts, not 27 independent scenes. Nano uses matched policy-sampling seeds. DreamZero-s2 fixes its noise seed, so repeats are not independent model-noise draws. We report models separately.

Measurements. Let $y_c(t)$ be the cube–bowl lateral offset for command c at recorded action t, positive to the robot's left. At a common action, the instruction response is

$$R(t) = y_{\text{LEFT}}(t) - y_{\text{RIGHT}}(t). \tag{1}$$

At separate terminal actions T_L, T_R, $R_{\text{end}} = y_{\text{LEFT}}(T_L) - y_{\text{RIGHT}}(T_R)$. Positive response means LEFT places the cube farther left relative to the bowl than RIGHT. Because the bowl moves, this is a relational measure, not absolute cube-endpoint separation.

The placement depth is the final offset toward the requested side: y_{LEFT} for LEFT and $-y_{\text{RIGHT}}$ for RIGHT. A larger positive depth means the cube finishes farther past the bowl on that side. We compare the two requests using

$$G = (-y_{\text{RIGHT}}) - y_{\text{LEFT}}. \tag{2}$$

Positive G favors rightward placement; negative G favors leftward placement. We measure its change between layouts within each trial block. Depth is a final position, not distance traveled from the start.

Task success requires release within 45 degrees of the requested direction from the bowl; lateral offset alone need not satisfy it. We use one rule in both layouts and measure all trials, including failures. These are executed simulator behaviors. Nano intervals use 20,000 bootstrap resamples of complete blocks; DreamZero-s2 is descriptive.

Both archived cohorts stop a success when the goal is first satisfied, while failures may reach the cap, so terminal positions are outcome-dependent. Only Nano retains stepwise positions. Its retrospective stopping check uses each pair's latest shared recorded action t_{pre} strictly before either trace terminates. This pair-specific action remains outcome-dependent, is not a common duration, and never carries a stopped trace forward. Action counts are simulator indices, not established physical time.

3 Results

Layout strongly changes DreamZero-s2 completion. DreamZero-s2 succeeds in 5/27 LEFT and 8/27 RIGHT trials originally, versus 25/27 each after reflection (Table 1). Completion rises from 13/54 to 50/54 with the same checkpoint, wording and controller. Original-scene failure is therefore insufficient evidence that direction words were ignored.

Table 1: Completion and instruction response differ. Success counts have 27 trials each. R_{end} compares unequal goal-stopped endpoints; G is achieved placement-depth difference, positive toward RIGHT.

Model	Layout	LEFT success	RIGHT success	R_{end} (cm)	G (cm)
Nano	Original	26/27	26/27	44.8	+14.8
	Reflected	27/27	23/27	45.3	−10.0
DreamZero-s2	Original	5/27	8/27	12.2	+2.3
	Reflected	25/27	25/27	37.6	−11.8

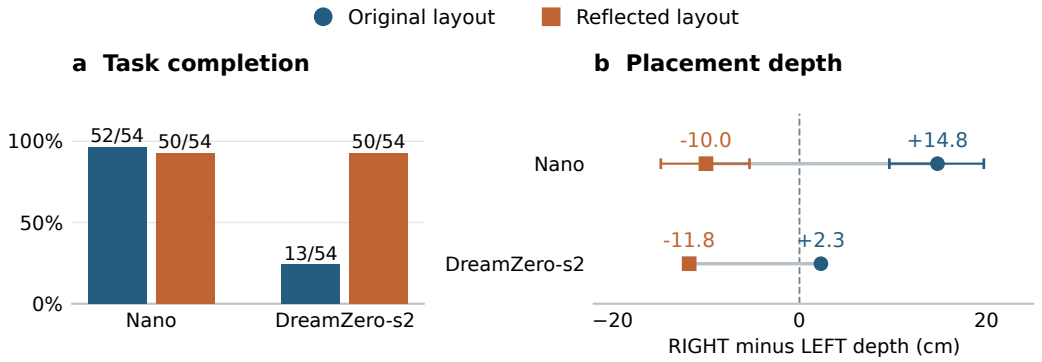


Figure 2: The intervention affects models differently. Left: successes out of 54. Right: mean depth gap G. Nano has 95% intervals over matched seed blocks; fixed-noise DreamZero-s2 is descriptive.

High success can hide changes in achieved placement. Nano completes 26/27 trials for each original-layout command, yet rightward depth exceeds leftward depth by 14.8 cm. After reflection, LEFT reaches 10.0 cm farther. The within-block change in G is −24.8 cm (95% interval $[-32.4, -17.3]$ cm over seed blocks). DreamZero-s2 changes from +2.3 to −11.8 cm (−14.1 cm). Figure 2 contrasts these shifts with completion.

Most original-layout DreamZero-s2 failures precede placement. Of its 41 unsuccessful trials, the recorded checks classify 26 as pickup failures, 14 as transport failures and one as wrong-side placement. These categories describe execution stages, rather than what the model understood. They show why the overall failure count cannot be treated as a count of directional-language errors.

Instruction response and placement depth need not change together. At unequal goal-stopped endpoints, Nano orders 27/27 pairs in both layouts with mean R_{end} 44.8 and 45.3 cm; DreamZero-s2 orders 26/27 and 27/27. Nano's latest shared strictly pretermination action retains 27/27 ordering and mean response 31.1 and 35.8 cm; pair actions span 89–290 (median 133) and 96–320 (median 149). At action 64 all pairs remain, but all four mean cube displacements are below 0.99 mm. At action 96, ordering is 20/26 (six reversed) and 25/27 (one reversed, one tie). At action 128 only 18/27 and 17/27 are observable, bounding full-cohort ordering at 17/27–26/27 and 16/27–26/27. At action 450 no matched pair remains. This is not A5 fixed-duration evidence or forecast accuracy.

On the same t_{pre} pairs, define $\Delta y_o = y_{o,\text{LEFT}}(t_{\text{pre}}) - y_{o,\text{RIGHT}}(t_{\text{pre}})$. Then $R = \Delta y_{\text{cube}} - \Delta y_{\text{bowl}}$: mean cube/bowl components are 31.121/0.017 cm originally and 35.865/0.020 cm after reflection. Bowl components range from −1.538 to 4.921 mm across both layouts, so only the mean response is cube-dominated. This is retained H01 A7-style decomposition, not completed prospective A7. The presentation videos use the lowest released H01 seed

(9400), selected without outcomes; the selected block happened to contain four successes while full H01 has six failures. They are not timing maps or outcome coverage.

First action-index requested-cone counts are 27/27 original LEFT, 27/27 original RIGHT, 27/27 reflected LEFT and 23/27 reflected RIGHT; all four absent entries are cap-censored reflected RIGHT trials. Restricted means (450 is a cap, not an event) are 168.2, 176.9, 119.6 and 209.6; the matched interaction has mean 81.4 actions (median 110; 20/27 positive). There are 91 sustained and 13 transient entries, and 104 entries but 102 successes; no common three-sample follow-up is imputed, so entry is not detached-release success. This diagnostic is neither physical-time latency, language understanding, A5–A7, nor forecast accuracy.

H05 dose response. The seven fixed bowl-y levels span 18 cm within 15 matched seed blocks (210 episodes). The registered within-seed RIGHT-minus-LEFT depth slope was 1.125 m/m (95% interval 0.719–1.562; 13/15 positive, $p = 0.0074$). Level means were not monotone; crossing 0.0188 m was outside the registered 0.0366–0.2166 m support. This fixed-scene result is not new-scene generalization or forecast accuracy.

H06 graded object-layout symmetry. Four inventory-matched positive levels (27 seeds; 216 episodes) yield a mean within-seed depth-gap slope of 0.020 m per unit realised asymmetry (95% interval 0.0004–0.0400; 20/27 positive, exact $p = 0.0192$). Inventory differs between level zero and the positive levels, so level zero is excluded. This is executed object-layout evidence, not position reflection, forecast accuracy, or robot/camera/embodiment symmetry.

4 Discussion and Limitations

Response, depth and completion differ; scene difficulty can mimic a spatial limit. H05 repeats seven positions in one setup, not seven independent scenes; H06 repeats positive levels within 27 blocks but has a different level-zero inventory. Neither isolates mechanisms. H01 stopping and outcome-blind videos are not fixed-duration A5, timing maps or forecast accuracy. LEFT/RIGHT is not general language comprehension. Without a non-world-model policy there is no causal-effect estimate; execution is not generated-video accuracy.

5 Conclusion

Changing layout alters spatial performance without changing models or commands. Separating response, depth and completion shows what success rate misses.

References

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